WIRESHARK NETWORK TRAFFIC ANALYSIS AND SOC INVESTIGATION

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Role: Cybersecurity Analyst Project

Tools: Wireshark, Ubuntu Linux, VirtualBox, Firefox

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1. Introduction

Network traffic analysis is a critical task performed by Security Operations Centre (SOC) analysts to monitor network activity, investigate communications, identify suspicious behaviour, and verify secure data transmission.

This project demonstrates the use of Wireshark to capture and analyse live network traffic generated within a Linux virtual machine. Multiple protocols were investigated to understand communication patterns and perform a SOC-style investigation.

2. Project Objectives

The objectives of this project were:

- Capture live network traffic using Wireshark.
- Analyse network communications across different protocol layers.
- Identify source and destination hosts.
- Examine DNS resolutions and website access.
- Investigate encrypted and unencrypted communications.
- Generate traffic statistics and endpoint information.
- Perform SOC investigation activities.

3. Lab Environment

Component	**Description**
Operating System	Ubuntu Linux
Virtualization Platform	VirtualBox
Packet Analyzer	Wireshark 4.6.4
Browser	Firefox
Interface Captured	enp0s3

4. Environment Validation

Prior to packet capture, system configuration and connectivity were validated.

Activities performed:

- Verified current user.
- Verified hostname.
- Verified network configuration.
- Confirmed internet connectivity using ICMP ping.

Figures:

ss

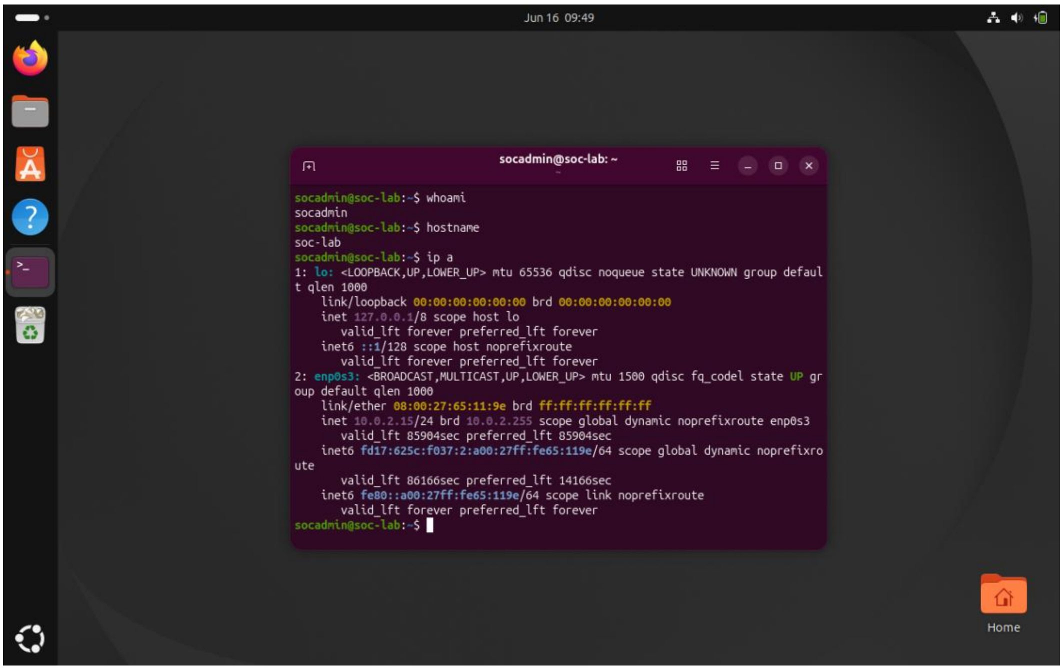

Figure 2 – Network Connectivity Validation

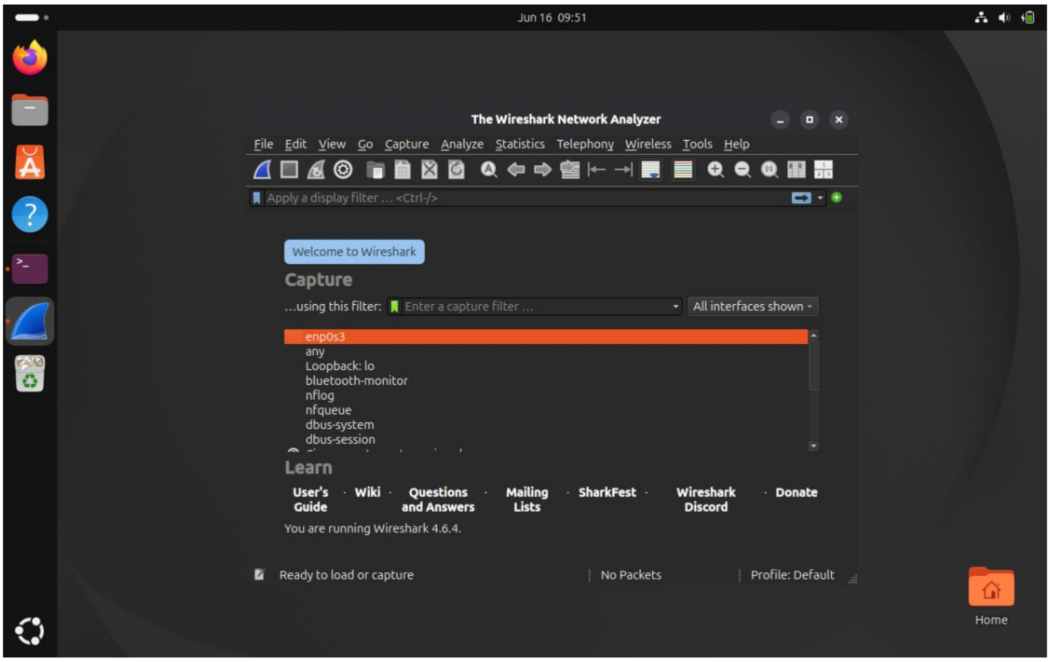

Figure 3 – Wireshark Interface Selection

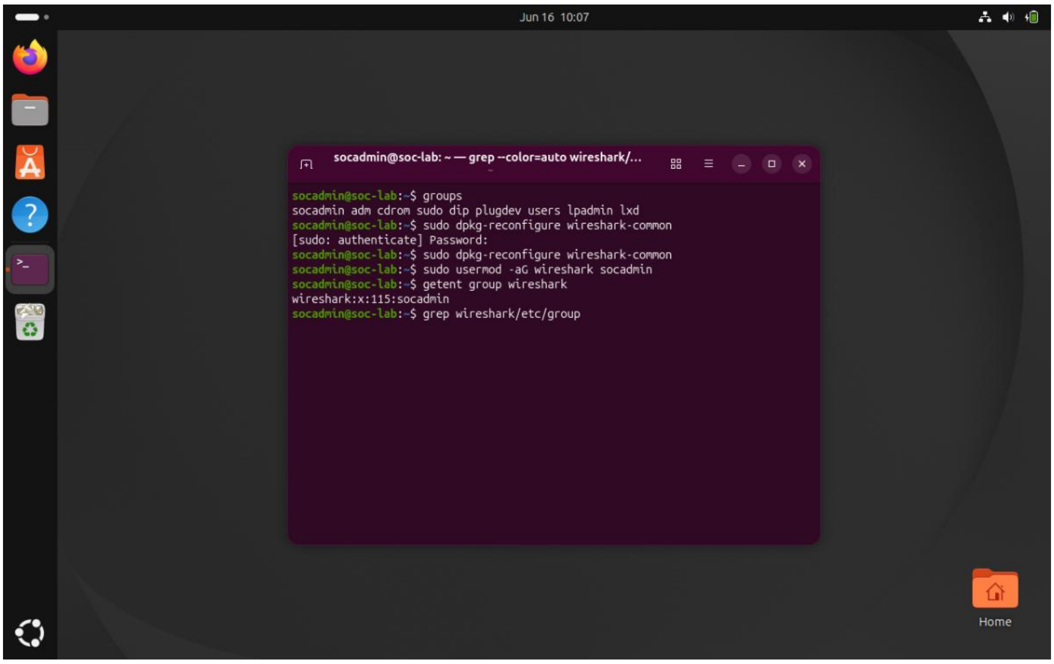

Figure 4 – Wireshark Permission Validation

5. Packet Capture

The network interface enp0s3 was selected as the active capture interface. Wireshark was configured to monitor live traffic while various web browsing and network activities were performed.

Traffic generated included:

- ICMP traffic
- DNS queries
- HTTP requests
- HTTPS/TLS communications
- TCP sessions

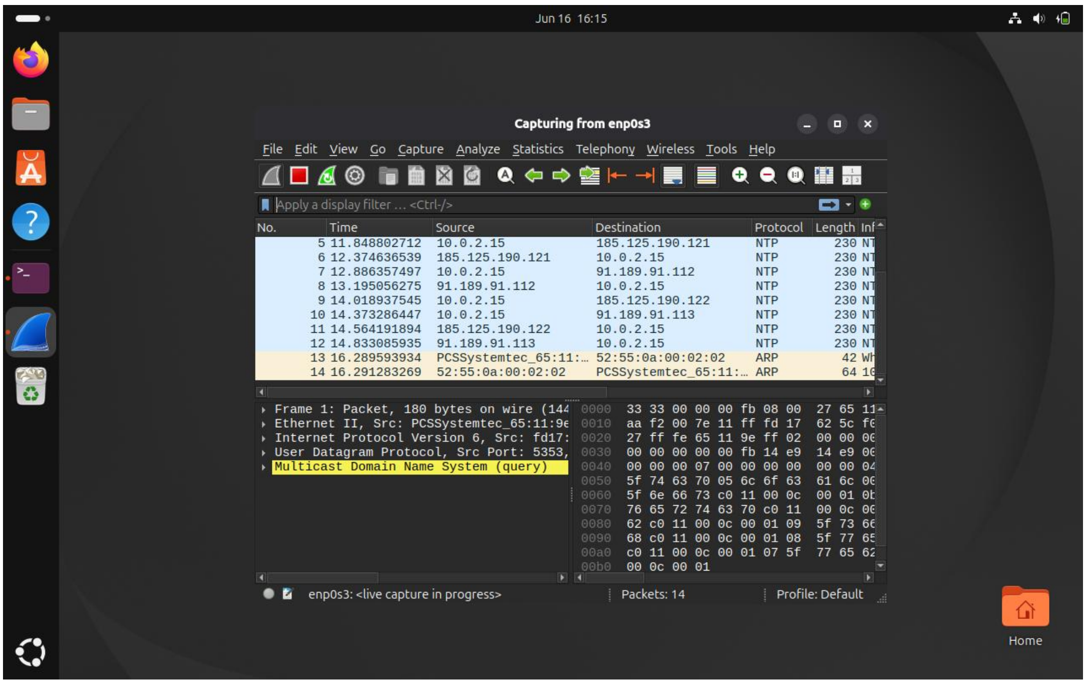

Figure 5 – Live Packet Capture Started

6. Protocol Analysis

6.1 ICMP Analysis

Display Filter:

icmp

ICMP packets were captured during connectivity testing. Echo Requests and Echo Replies were observed, confirming successful communication between the host and external systems.

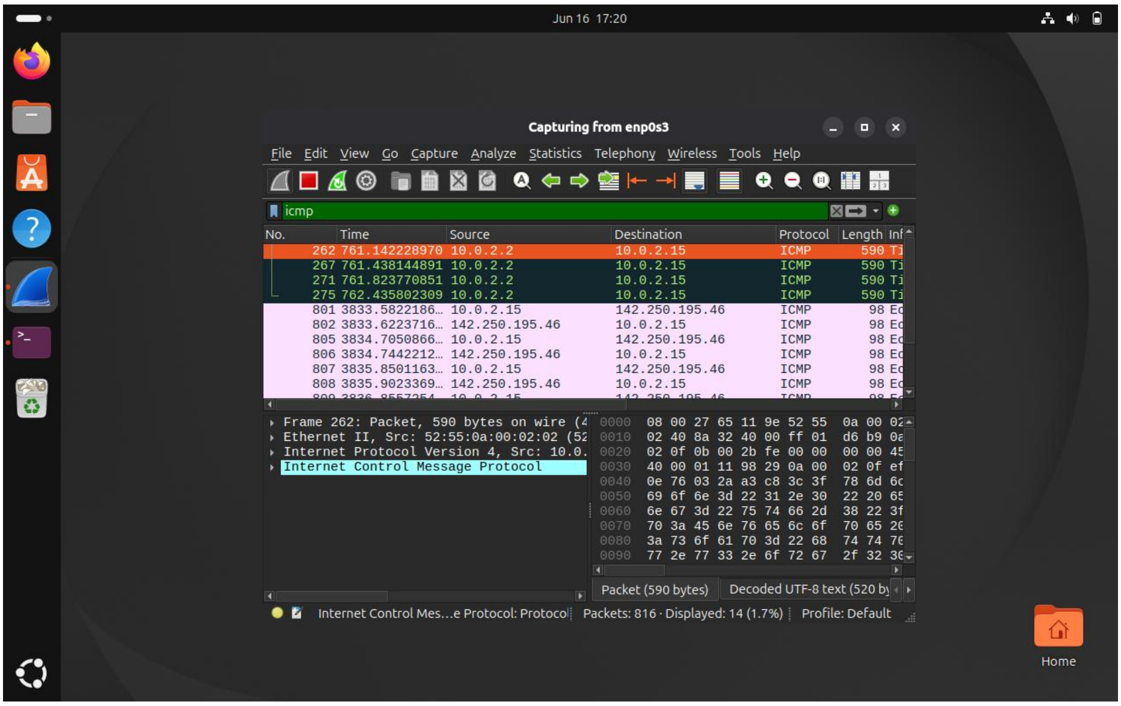

Figure 6 – ICMP Traffic Analysis

6.2 DNS Analysis

Display Filter:

dns

DNS traffic was analyzed to identify domain name resolution activity. Queries and responses revealed communication between the client and DNS server.

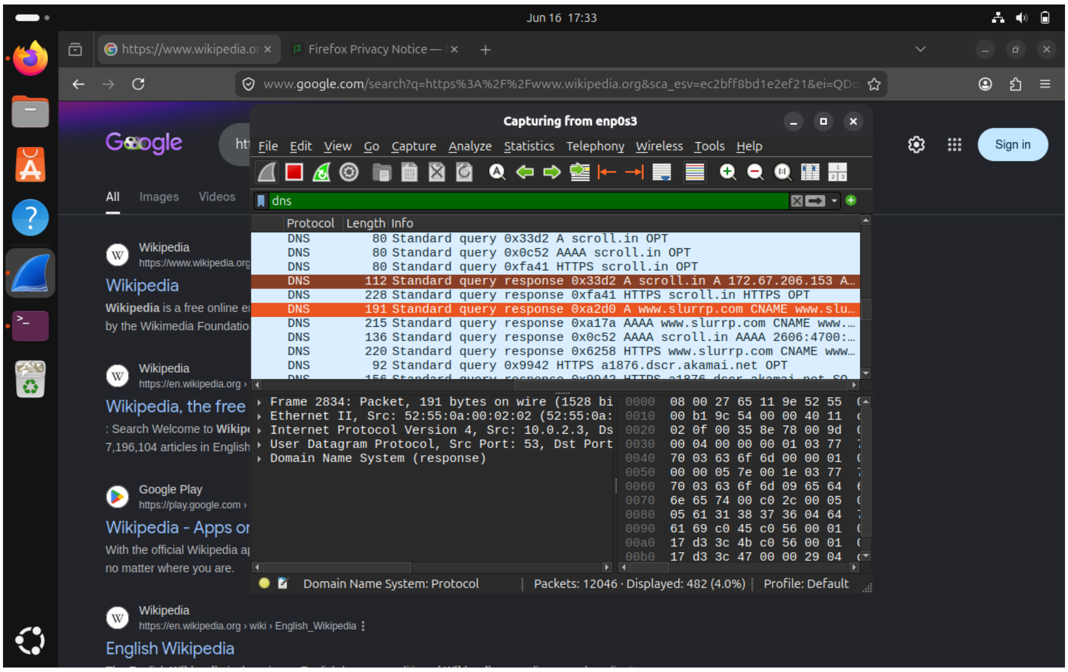

Figure 7 – DNS Traffic Analysis

6.3 HTTP Analysis

Display Filter:

http

Unencrypted web traffic was analyzed using the NeverSSL website. HTTP GET requests and responses were visible in plaintext.

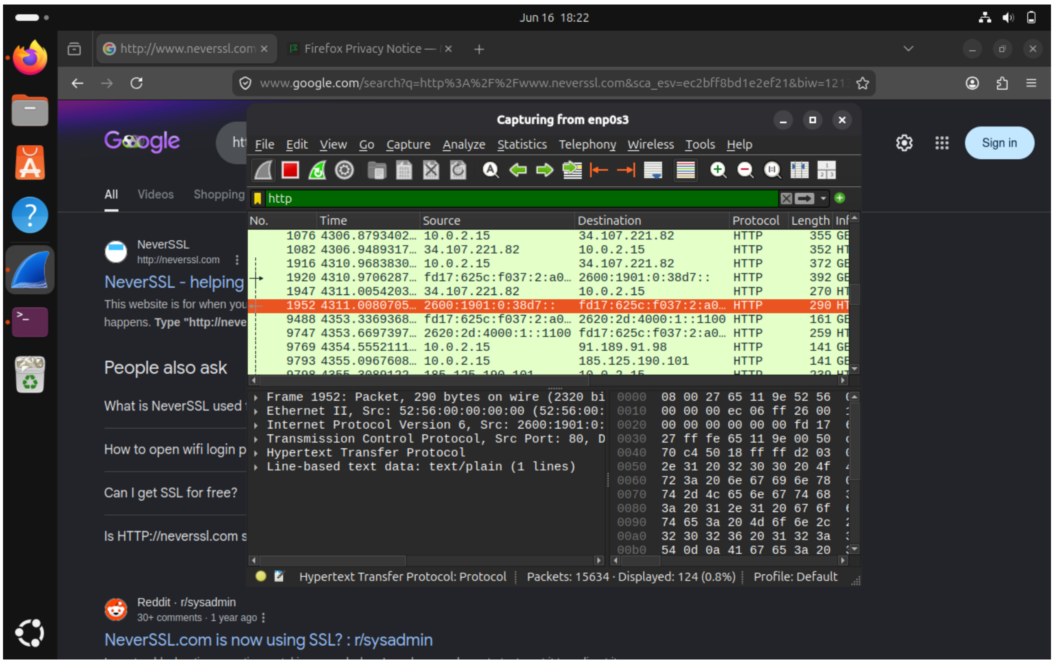

Figure 8 – HTTP Traffic Analysis

6.4 TLS Analysis

Display Filter:

tls

Encrypted HTTPS communications were captured and analyzed. TLS 1.2 and TLS 1.3 sessions were observed, demonstrating secure web communication.

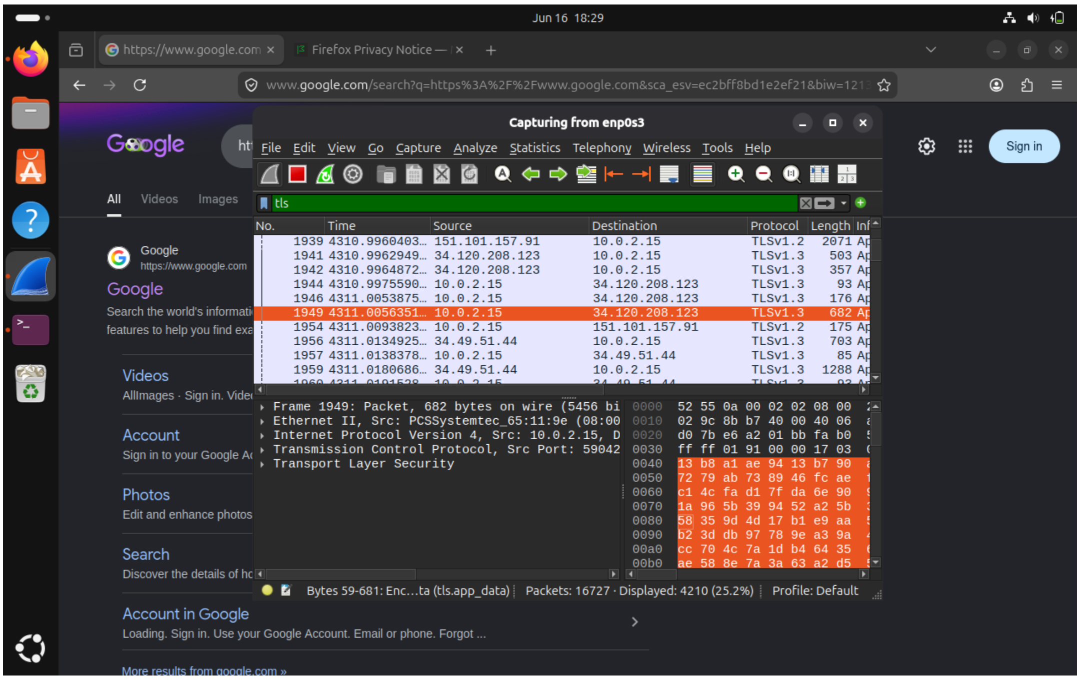

Figure 9 – TLS Traffic Analysis

6.5 TCP Analysis

Display Filter:

tcp

TCP traffic analysis revealed reliable communication sessions between client and server hosts. Packet exchanges demonstrated transport-layer communication over established sessions.

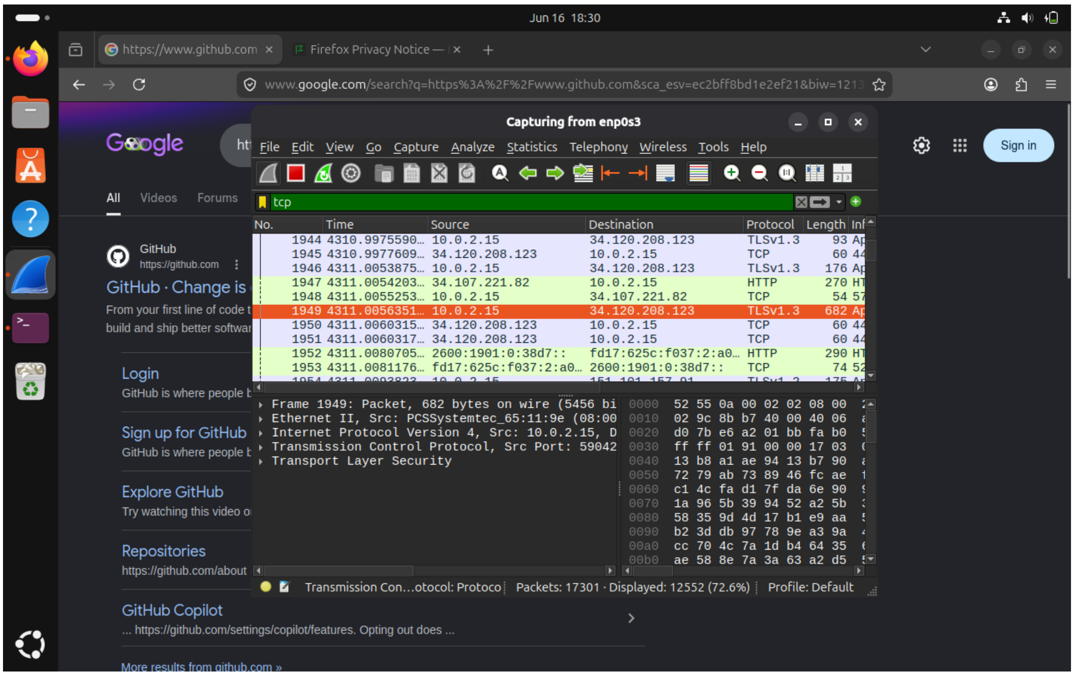

Figure 10 – TCP Traffic Analysis

7. Traffic Statistics

Endpoint Statistics

Wireshark Endpoint Statistics were used to identify active hosts participating in communications.

Most active endpoints included:

- 10.0.2.15
- 34.49.51.44
- 151.101.157.91

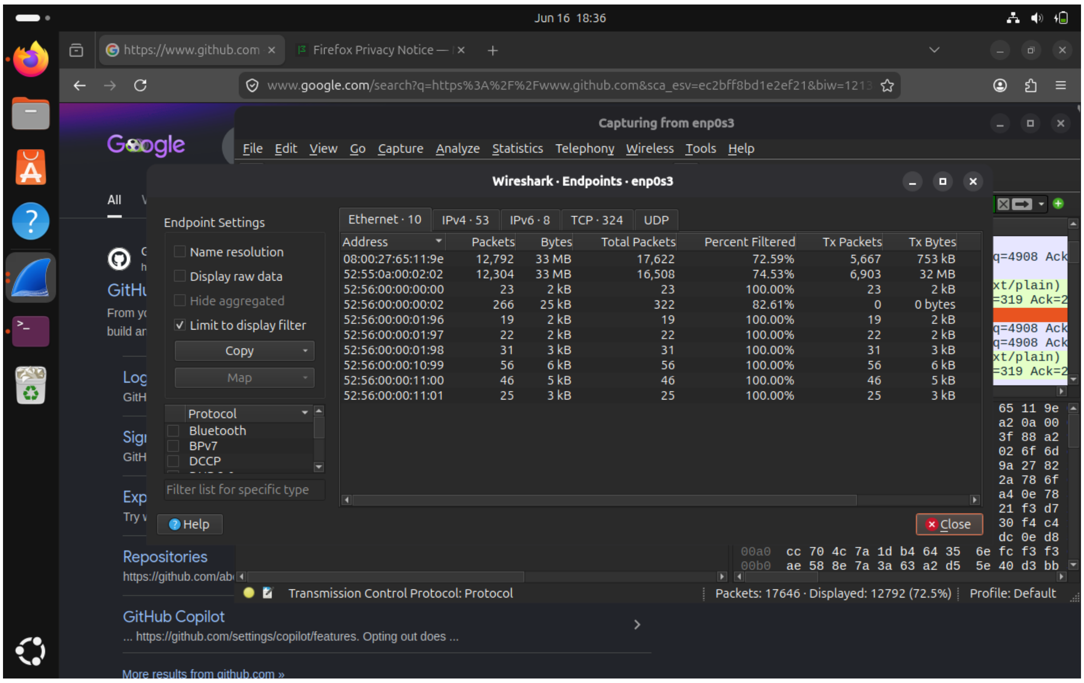

Figure 11 – Endpoint Statistics

Protocol Hierarchy Statistics

Protocol Hierarchy Statistics were generated to identify traffic distribution across network protocols.

Observed traffic distribution:

- IPv4: 96.2%
- IPv6: 3.8%
- TCP: 96.2%
- TLS: 33.9%
- HTTP: 0.5%

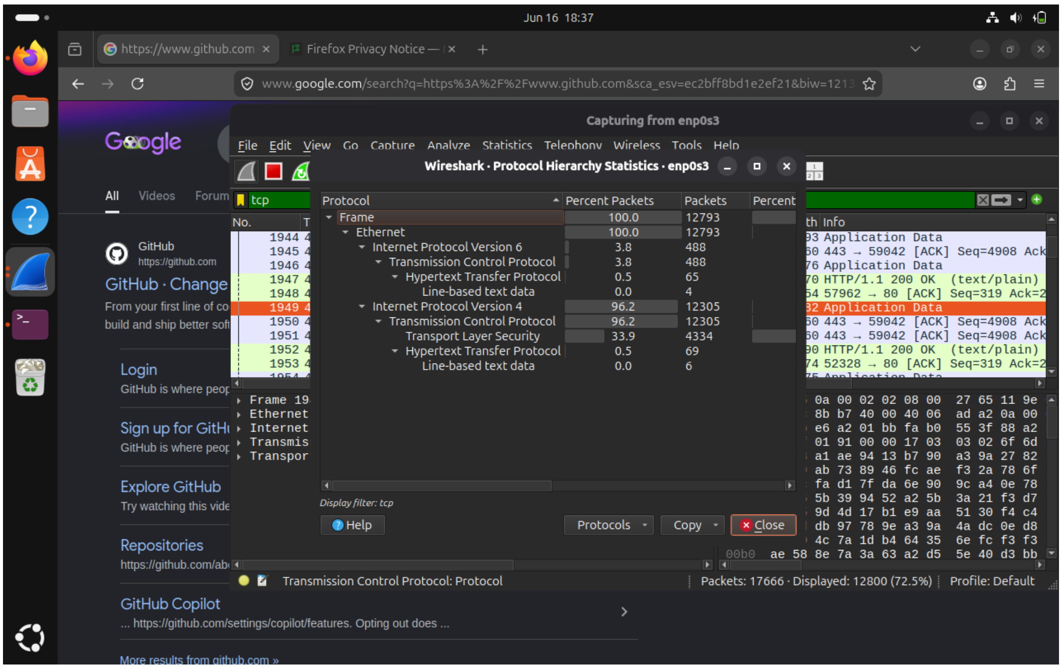

Figure 12 – Protocol Hierarchy Statistics

8. SOC Investigation

A basic SOC investigation was performed using DNS, TCP, and TLS packet analysis.

Question 1: What domain was visited?

Observed domains:

- github.com
- google.com
- wikipedia.org
- neverssl.com

Question 2: What IP addresses were returned?

Observed IP addresses included:

- 34.49.51.44
- 151.101.157.91

Additional addresses were identified through DNS responses.

Question 3: What protocols were used?

Protocols identified:

- DNS
- ICMP
- HTTP
- TCP
- TLS

Question 4: Was communication encrypted?

Yes.

TLS 1.2 and TLS 1.3 traffic was observed on TCP port 443, confirming encrypted HTTPS communication.

Figures:

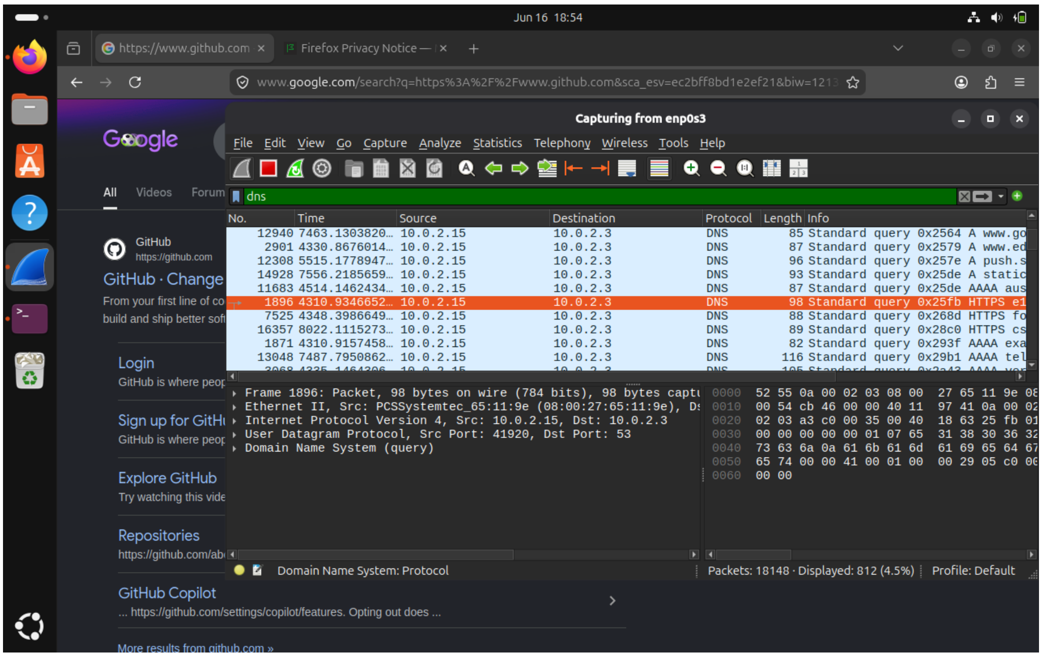

Figure 13 – SOC DNS Investigation

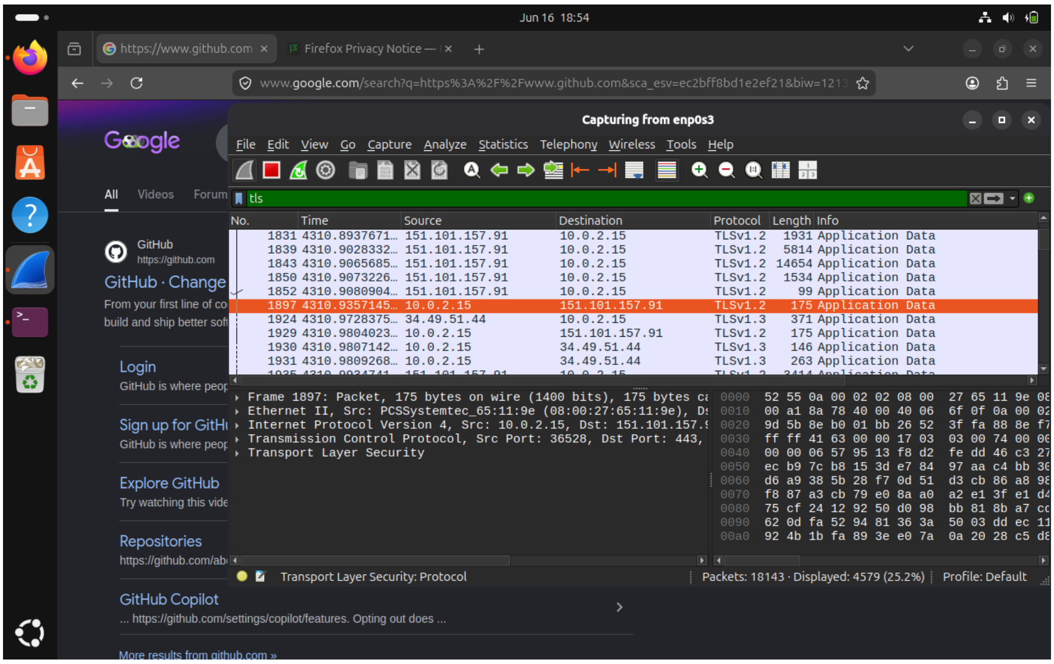

Figure 14 – SOC TLS Investigation

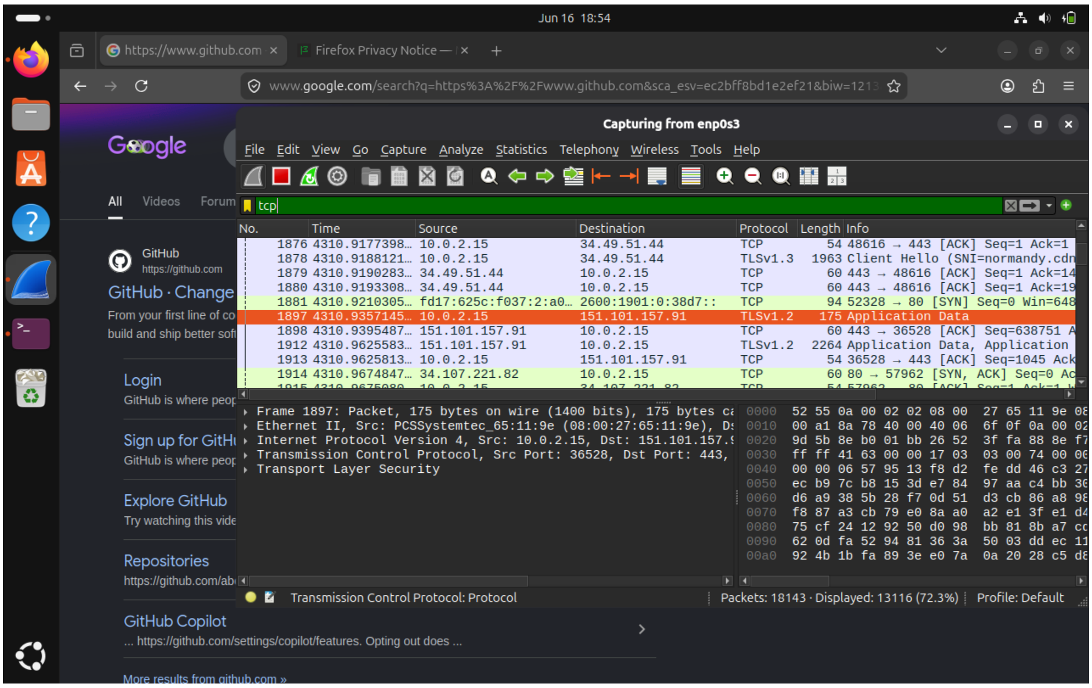

Figure 15 – SOC TCP Session Analysis

9. Findings

The analysis confirmed successful capture and investigation of network communications. DNS queries were used to resolve domain names, HTTP traffic demonstrated unencrypted communications, and TLS sessions confirmed encrypted HTTPS traffic.

Endpoint and protocol statistics provided visibility into network activity, enabling effective monitoring and investigation.

10. Conclusion

This project successfully demonstrated network traffic capture, protocol analysis, and SOC investigation using Wireshark. Practical experience was gained in monitoring ICMP, DNS, HTTP, TLS, and TCP communications while developing skills relevant to Security Operations Center (SOC) analyst responsibilities.

Key skills developed include:

- Packet Analysis
- Network Monitoring
- Traffic Investigation
- Protocol Analysis
- DNS Analysis
- Endpoint Monitoring
- Wireshark Operations
- SOC Investigation Techniques